

Symmetry in three dimensions

Cambridge insert: planes of symmetry and axes of rotational symmetry, on the solids the rest of the year is built from.

LESSON 3

1 × 40 MIN

GEOMETRY 10, §2 (EXTENSION)

IGCSE E4.6

LESSON CLOCK

8'

14'

12'

8'

3'

Plane versus axis	8 min
Counting the planes of a cuboid	14 min
Rotational axes and their order	12 min
Using symmetry to save work	8 min
Homework	3 min

LEARNING OBJECTIVES

- Find the planes of symmetry of a cuboid, prism, pyramid, cylinder and cone.
- Find the axes of rotational symmetry and state each order.
- Distinguish a plane of symmetry from an axis of symmetry.
- Use symmetry to shorten a calculation.

TEXTBOOK

National	pp. 15–18
Cambridge	Core & Extended, pp. 268–273
Workbook	IGCSE Exercise 4.6

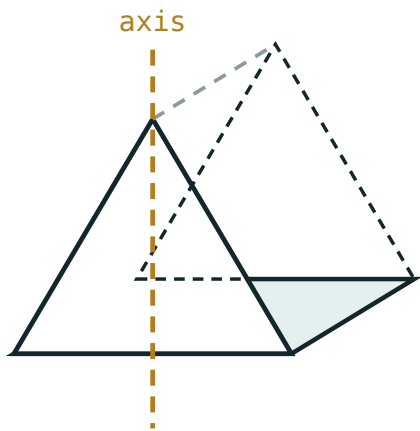
01

Explanation

Two different things

PLANE AND AXIS

A **plane of symmetry** slices the solid into two mirror halves. An **axis of rotational symmetry** is a line about which the solid can be turned by less than a full turn and look unchanged; the number of positions is the **order**.



a plane of symmetry cuts it into mirror halves

*A cuboid cut by one of its three planes of symmetry,
and turned about one of its axes.*

A solid can have many of each, and the two counts are independent — a cone has infinitely many planes but exactly one axis.

Counting them

SOLID	PLANES	AXES (WITH ORDER)
cuboid, all edges different	3	3, each of order 2
square-based cuboid	5	1 of order 4, 4 of order 2
cube	9	3 of order 4, 4 of order 3, 6 of order 2
triangular prism (equilateral)	4	1 of order 3, 3 of order 2
square-based pyramid	4	1 of order 4
cylinder	infinitely many, plus 1 horizontal	1 of infinite order
cone	infinitely many	1 of infinite order
sphere	infinitely many	infinitely many

THE CUBOID TRAP

A cuboid has **3** planes of symmetry, not 6. The two halves either side of a plane must be mirror images — a diagonal cut through a rectangular face does not give that.

Symmetry as a shortcut

Symmetry is not decoration. It halves work:

- The centre of mass lies on every plane of symmetry — so on their intersection.
- A section perpendicular to an axis of order n repeats n times.
- Two lengths symmetric about a plane are equal, with no calculation.

In a cube of side a , the two space diagonals from one face meet on the vertical axis, so their intersection is at height $\frac{a}{2}$ — read straight from symmetry.

02 Worked examples

EXAMPLE 1 How many planes of symmetry has a cuboid $3 \times 4 \times 5$?

- 1 All three edge lengths differ.
No diagonal plane works.
- 2 One plane parallel to each pair of opposite faces.

ANSWER

3

EXAMPLE 2 A prism has a regular hexagonal cross-section. How many planes of symmetry?

① The hexagon has 6 lines of symmetry.
6 vertical planes.

② One horizontal plane halfway up.

ANSWER

7

EXAMPLE 3 State the order of rotational symmetry of a square-based pyramid about its vertical axis.

① Turning by 90° maps the square base to itself.

② Four positions in a full turn.

ANSWER

Order 4

03 Interactive model

Hold a cuboid box and slice it with a sheet of card to show each plane.

Counting symmetry

A cuboid with all edges different has how many planes of symmetry?

A cube has how many planes of symmetry?

A cone has how many axes of rotational symmetry?

A square-based pyramid has rotational order:

1

2

4

8

A regular triangular prism has how many planes?

3

4

5

6

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Plane of symmetry	Simmetriya tekisligi	Плоскость симметрии
Axis of symmetry	Simmetriya o'qi	Ось симметрии
Rotational symmetry	Aylanma simmetriya	Вращательная симметрия
Order of symmetry	Simmetriya tartibi	Порядок симметрии
Reflection	Akslantirish	Отражение
Mirror image	Ko'zgu tasviri	Зеркальное изображение
Cuboid	Parallelepiped	Прямоугольный параллелепипед
Cross-section	Kesim	Сечение
Regular solid	Muntazam jism	Правильное тело

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

A plane of symmetry produces:

two mirror halves

two equal volumes only

a rotation

a shadow

The order of rotational symmetry of a cylinder about its axis is:

1

2

4

infinite

A sphere has how many planes of symmetry?

1

3

9

infinitely many

A cube has axes of order 3 through:

face centres

opposite vertices

edge midpoints

it has none

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Planes of symmetry of a cube

ANSWER 9

2. Planes of symmetry of a $2 \times 3 \times 4$ cuboid

ANSWER 3

3. Planes of symmetry of a sphere

ANSWER infinitely many

4. Rotational order of a cube about a face axis

ANSWER 4

5. Axes of a cone

ANSWER 1

6. Planes of a square-based pyramid

ANSWER 4

7. Rotational order of a cylinder

ANSWER infinite

Medium · the standard question

7 PROBLEMS

1. Planes of a regular hexagonal prism

ANSWER 7

2. Planes of a regular pentagonal pyramid

ANSWER 5

3. Rotational order of a regular hexagonal prism about its long axis

ANSWER 6

4. Planes of a cuboid with a square base $3 \times 3 \times 5$

ANSWER 5

5. Axes of order 2 in a cube

ANSWER 6

6. Axes of order 3 in a cube

ANSWER 4

7. Total rotational axes of a cube

ANSWER 13

Hard · stretch and reason

7 PROBLEMS

1. Explain why a cuboid $a \times a \times b$ with $a \neq b$ has 5 planes

ANSWER 3 face planes plus 2 diagonal planes of the square cross-section

2. A regular tetrahedron: count its planes

ANSWER 6

3. A regular tetrahedron: count its rotational axes and orders

ANSWER 4 of order 3, 3 of order 2

4. Why has a cone infinitely many planes but only one axis?

ANSWER Every plane through the axis is a mirror; only that line is fixed

5. A solid has 1 plane only. Sketch one

ANSWER e.g. a scalene-triangle prism cut asymmetrically

6. Use symmetry to find the centre of a cube of side 6 from a corner

ANSWER (3, 3, 3)

7. A cuboid $a \times a \times a$ — why does the count jump from 5 to 9?

ANSWER The three square cross-sections each contribute two diagonal planes

07 Homework

Homework — 4 tasks

Sketch each solid once and draw the planes on the sketch.

- Count the planes of symmetry of a cube, a $2 \times 2 \times 5$ cuboid and a regular hexagonal prism.
- State every rotational axis of a cube with its order, and check the total is 13.
- Find a solid with exactly two planes of symmetry and sketch it.
- Explain why a plane of symmetry always divides a solid into equal volumes, but equal volumes do not imply a plane of symmetry.